

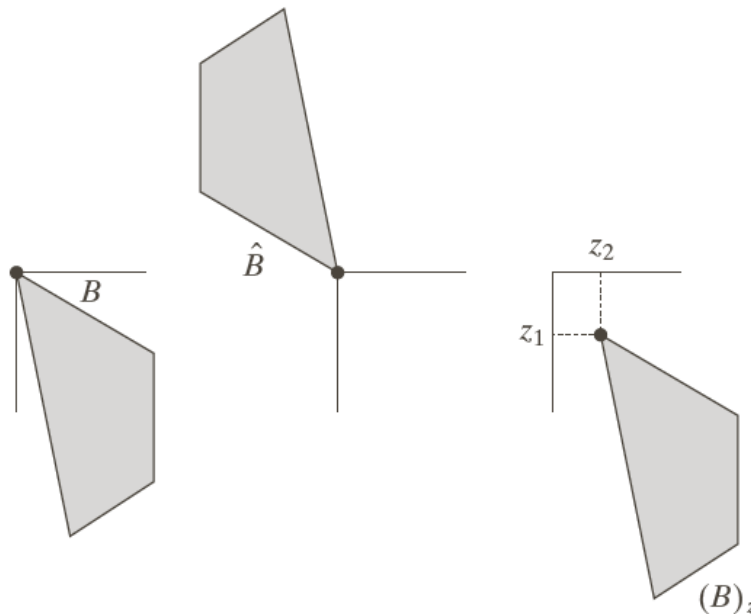


Chapter 9

Morphological Image Processing

9.1 Preliminaries

- Reflection of set B $\hat{B} = \{w \mid w = -b, b \in B\}$
- Translation of set B $(B)_z = \{c \mid c = b + z, b \in B\}$



a b c

FIGURE 9.1

(a) A set, (b) its reflection, and (c) its translation by z .

Structuring Elements

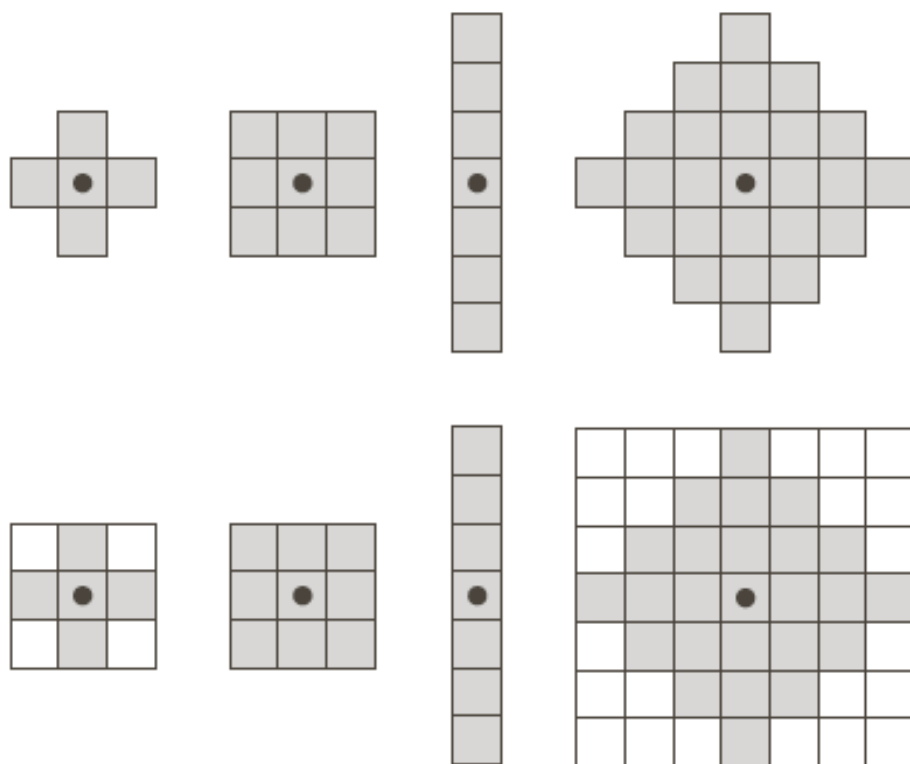


FIGURE 9.2 First row: Examples of structuring elements. Second row: Structuring elements converted to rectangular arrays. The dots denote the centers of the SEs.

9.2 Erosion and Dilation

- 9.2.1 Erosion

$$A(-)B = \{z \mid (B)z \subset A\}$$

$$A(-)B = \{z \mid (B)z \cap A^c = \phi\}$$

9.2.1 Erosion

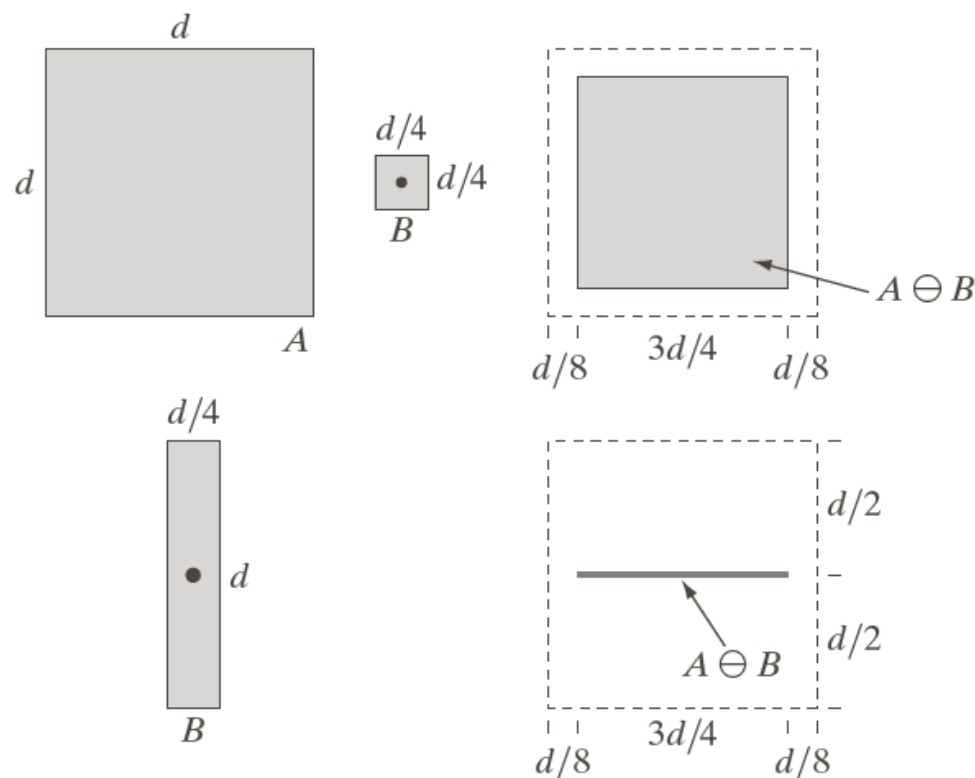
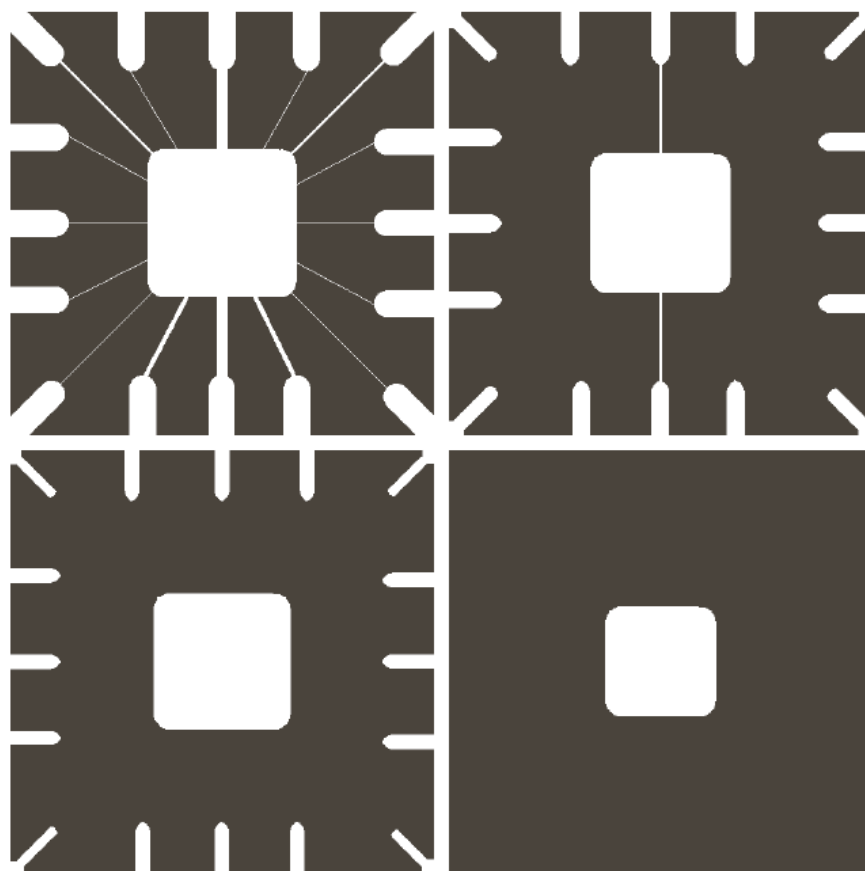


FIGURE 9.4 (a) Set A . (b) Square structuring element, B . (c) Erosion of A by B , shown shaded. (d) Elongated structuring element. (e) Erosion of A by B using this element. The dotted border in (c) and (e) is the boundary of set A , shown only for reference.

9.2.1 Erosion

- [Programming]

9.2.1 Erosion



a	b
c	d

FIGURE 9.5 Using erosion to remove image components. (a) A 486×486 binary image of a wire-bond mask. (b)–(d) Image eroded using square structuring elements of sizes 11×11 , 15×15 , and 45×45 , respectively. The elements of the SEs were all 1s.

9.2.1 Erosion

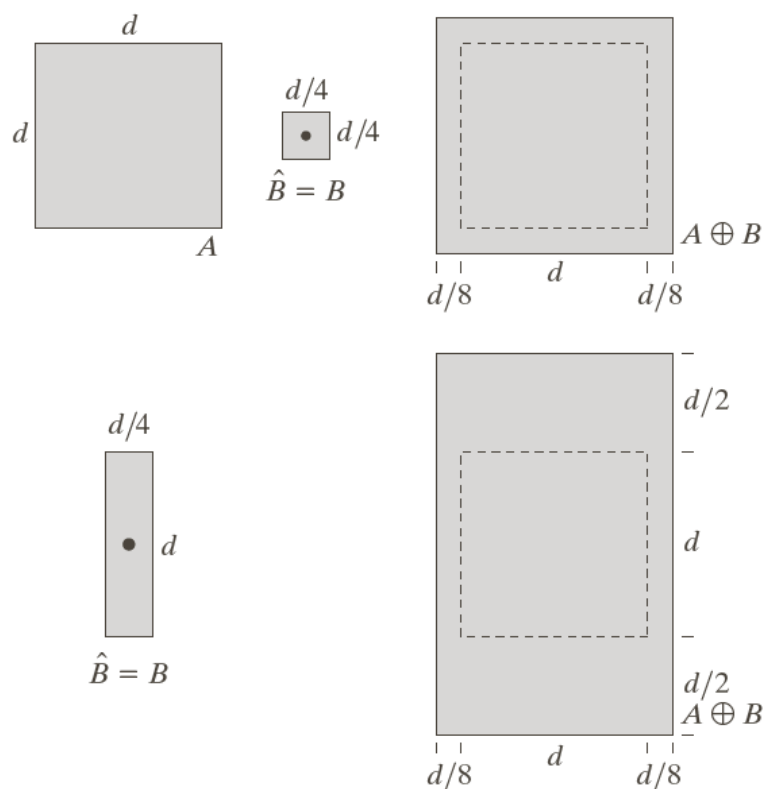
- Result of Erosion
 - Shrink
 - Smooth Boundary
 - Remove thin lines and small region

9.2 Erosion and Dilation

- 9.2.2 Dilation

$$A(+)B = \{z \mid (B)z \cap A \neq \phi\}$$

9.2.2 Dilation



a	b	c
d		e

FIGURE 9.6

(a) Set A .
 (b) Square structuring element (the dot denotes the origin).
 (c) Dilation of A by B , shown shaded.
 (d) Elongated structuring element.
 (e) Dilation of A using this element. The dotted border in (c) and (e) is the boundary of set A , shown only for reference

9.2.2 Dilation

- [Programming]

9.2.2 Dilation

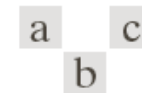
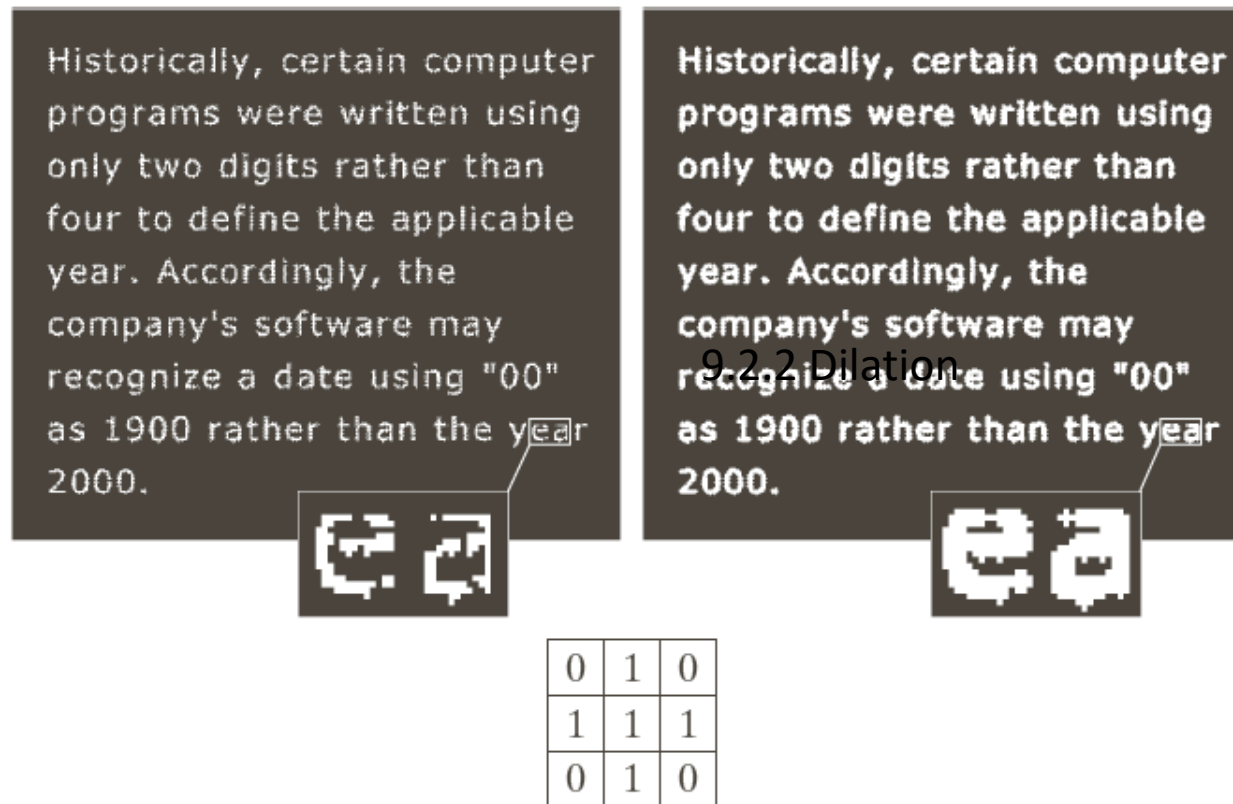


FIGURE 9.7

(a) Sample text of poor resolution with broken characters (see magnified view).
 (b) Structuring element.
 (c) Dilation of (a) by (b). Broken segments were joined.

9.2.2 Dilation

- Result of Dilation
 - Expand
 - Smooth Boundary
 - Remove thin rift and small hole

9.3 Open and Close

- Open: $A(-)B(+)B$
- Close: $A(+)B(-)B$

9.3 Open and Close

- Convex corner

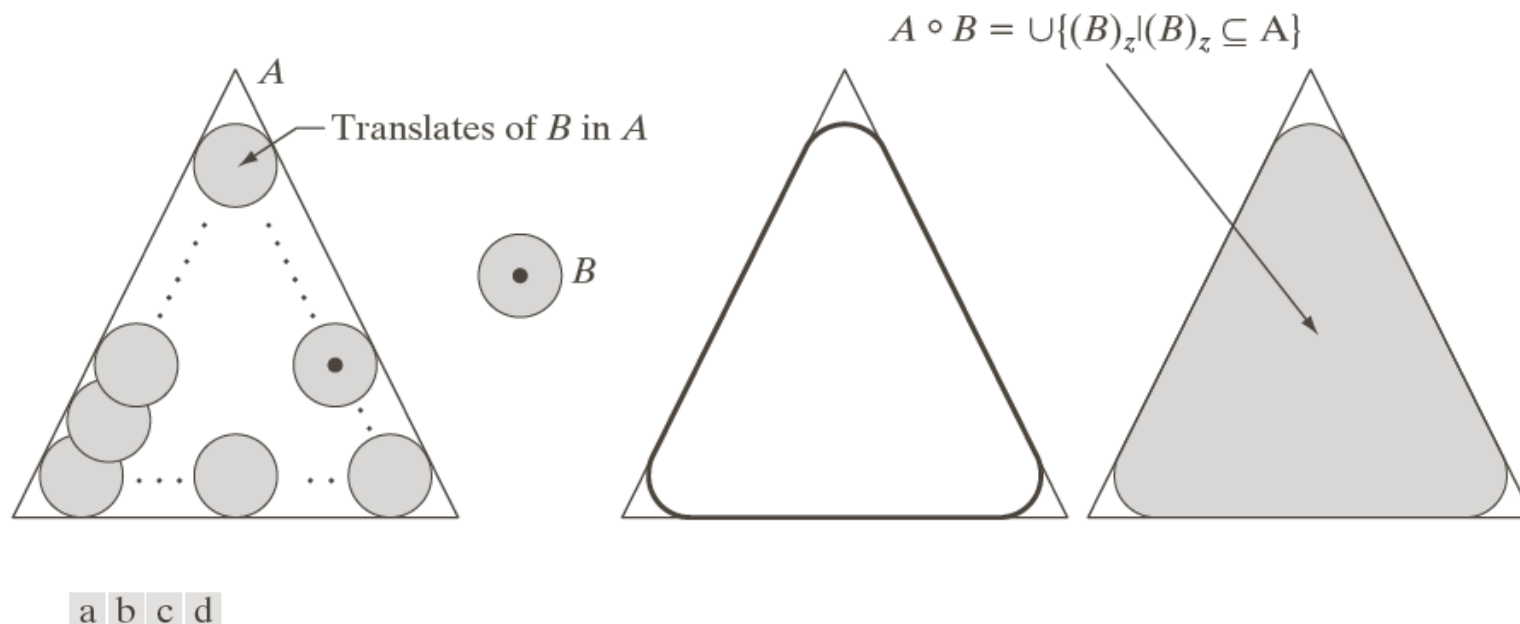


FIGURE 9.8 (a) Structuring element B “rolling” along the inner boundary of A (the dot indicates the origin of B). (b) Structuring element. (c) The heavy line is the outer boundary of the opening. (d) Complete opening (shaded). We did not shade A in (a) for clarity.

9.3 Open and Close

- Concave corner

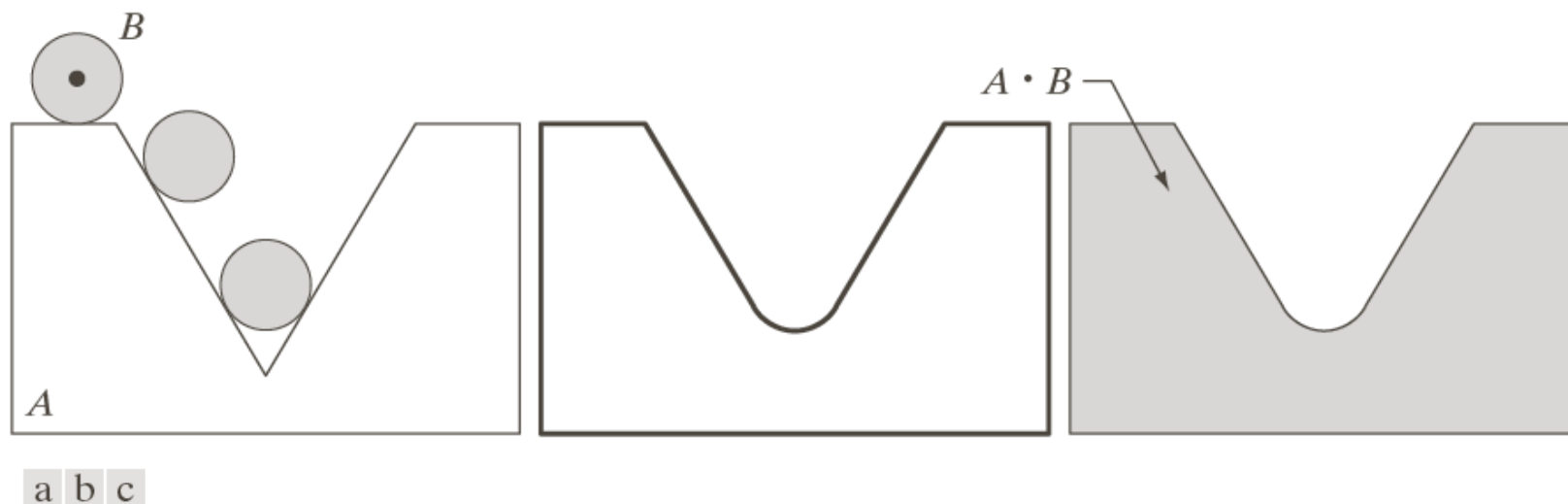


FIGURE 9.9 (a) Structuring element B “rolling” on the outer boundary of set A . (b) The heavy line is the outer boundary of the closing. (c) Complete closing (shaded). We did not shade A in (a) for clarity.

9.3 Open and Close

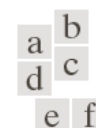


FIGURE 9.11

(a) Noisy image.
 (b) Structuring element.
 (c) Eroded image.
 (d) Opening of A .
 (e) Dilation of the opening.
 (f) Closing of the opening.
 (Original image courtesy of the National Institute of Standards and Technology.)